

## 0.1 Abel's Formular and Abel's Test

### 0.1.1 Abel's Formular: Summation by Parts

**Lemma 0.1.1.** Given sequences  $\{a_n\}$  and  $\{b_n\}$ , denote  $A_n = \sum_{k=0}^n a_k$ .  
Then, for any  $q \in \mathbb{N}$ ,

$$\sum_{n=0}^q a_n b_n = \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left( \sum_{k=0}^n a_k \right) \cdot (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^q a_n.$$

In particular, if  $0 \leq p \leq q$ ,

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

*Proof.* Since  $a_n = A_n - A_{n-1}$  (with  $A_{-1} \stackrel{\text{def}}{=} 0$ ),

$$\begin{aligned} \sum_{n=0}^q a_n b_n &= \sum_{n=0}^q (A_n - A_{n-1}) b_n = \sum_{n=0}^q A_n b_n - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + \sum_{n=0}^q A_n b_{n+1} - \sum_{n=0}^q A_{n-1} b_n \\ &= \sum_{n=0}^q A_n (b_n - b_{n+1}) + A_q b_{q+1} \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q. \end{aligned}$$

Particular,

$$\begin{aligned} \sum_{n=p}^q a_n b_n &= \sum_{n=0}^q a_n b_n - \sum_{n=0}^{p-1} a_n b_n \\ &= \sum_{n=0}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - \left( \sum_{n=0}^{p-2} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} \right) \\ &= \sum_{n=p-1}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1} \\ &= \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p. \end{aligned}$$

□

**Note:** Denote  $A_n = \sum_{k=0}^n a_k$  and  $\Delta b_n = b_{n+1} - b_n$ , then

$$\sum_{n=0}^q a_n b_n = A_q b_q - \sum_{n=0}^{q-1} A_n \Delta b_n$$

It is similar to:

$$\int f \cdot g = Fg - \int Fg'$$

### 0.1.2 Abel's Test

**Theorem 0.1.2.** If  $\sum a_n$  converges and  $\{b_n\}$  is bounded and monotone sequence, then  $\sum a_n b_n$  converges.

*Proof.* Since  $\{b_n\}$  is bounded and monotone, it is convergent. Put  $b = \lim_{n \rightarrow \infty} b_n$  and  $A_n = \sum_{k=0}^n a_k$ ,  $A_{-1} = 0$ . WLOG, suppose  $\{b_n\}$  is monotone decreasing. Then  $\{b_n - b\}$  converges to zero and is decreasing.

Let  $M > 0$  be a bound of  $|A_n|$ . Given  $\varepsilon > 0$ , choose  $N \in \mathbb{N}$  such that  $n \geq N \implies |b_n - b| < \frac{\varepsilon}{2M}$ .

By Abel's formula, for all  $q \geq p \geq N$ ,

$$\begin{aligned} \left| \sum_{n=p}^q a_n(b_n - b) \right| &= \left| \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q(b_q - b) - A_{p-1}(b_p - b) \right| \\ &\leq M \cdot \left[ \sum_{n=p}^{q-1} \underbrace{|b_n - b_{n+1}|}_{\geq 0} + \underbrace{|b_q - b|}_{\geq 0} + \underbrace{|b_p - b|}_{\geq 0} \right] \\ &= M \cdot \left[ \sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b) \right] \\ &= M \cdot [(b_p - b_q) + (b_q - b) + (b_p - b)] \\ &= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon \end{aligned}$$

Therefore, by the Cauchy Criterion for series,  $\sum a_n(b_n - b)$  converges.

Moreover, since  $\sum a_n$  converges, so is  $b \cdot \sum a_n$ . Finally,

$$\sum_{n=0}^{\infty} a_n(b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

converges as a sum of two convergent series. □

**Note:** If  $b = 0$ , it is sufficient for the convergence of  $\sum a_n b_n$  that  $\sum a_n$  be bounded.

In this case,  $\sum a_n$  does not necessarily have to converge (it is called Dirichlet's Test).

**Example. A Series**

$$\sum_{n=1}^{\infty} \frac{\sin n}{n}$$

converges. Enough to show that:  $\sum_{n=1}^N \sin n$  converges. Then we can apply Dirichlet's Test. Observe that

$$\begin{aligned} 2 \sin(1) \cdot \sum_{n=1}^N \sin n &= \sum_{n=1}^N 2 \sin(1) \sin n = \sum_{n=1}^N [\cos(n-1) - \cos(n+1)] \\ &= \cos 0 - \cos 2 + \cos 1 - \cos 3 + \cos 2 - \cos 4 + \cdots + \cos(N-2) - \cos(N) + \cos(N-1) + \cos(N+1) \\ &= \cos 0 + \cos 1 - \cos(N) - \cos(N+1) \end{aligned}$$

Therefore, for any  $N \in \mathbb{N}$ ,

$$\left| \sum_{n=1}^N \sin n \right| = \left| \frac{\cos 0 + \cos 1 - \cos(N) - \cos(N+1)}{2 \sin(1)} \right| \leq \frac{2}{\sin(1)}$$

Now, Applying the Dirichlet's Test, we obtain the result. But, it is not converges absolutely.